FINAL RESEARCH PROJECT - PROCESS REPORT

Storage-Safe Cutover Execution

From a confirmed quota failure to a rollback-safe release and a verified five-algorithm smoke run.

49.94%	248	5 / 5
OBSERVED OUTPUT REDUCTION	RELEASE FILES VERIFIED	ALGORITHMS COMPLETE

CUTOVER OUTCOME

The storage-safe release was applied and preflighted successfully. Job 3550 completed with exit 0:0. Rollback was not required. Full screening remains unsubmitted.

TECHNICAL SUMMARY

Job 3542 exposed a storage failure at checkpoint write time. The implemented response suppresses intermediate checkpoint writes before they occur, retains the final model and scientific outputs, and emits checksum-bound retention receipts. A rollback snapshot and accepted calibration-lineage receipt preserved the prior research boundary. The final release passed remote verification and one storage-safe smoke job.

Campaign	large_scale_thai_drl_v2
Evidence cut	17 July 2026
Generated	20 July 2026
Audience	Research team, technical reviewer, and HPC operator

Evidence basis: incident receipt, rollback snapshot, accepted lineage receipt, deterministic release manifest, remote preflight, Slurm accounting, and strict import validation.

SECTION 1

Failure mechanism and engineering response

CONFIRMED ROOT CAUSE

Job 3542 failed after 20 seconds with OSError: [Errno 122] Disk quota exceeded. New outputs inherited Unix group s00658; its Lustre soft-quota grace had expired even though project-ID quota 25110303 remained healthy.

Failure traced
Checkpoint zip close hit group quota

→

Writes suppressed
No intermediate checkpoint creation

→

Evidence retained
Final model, metrics, curves, actions

→

Receipts signed
Per-trial hash and policy evidence

→

Reuse guarded
Unsafe or incomplete reuse rejected

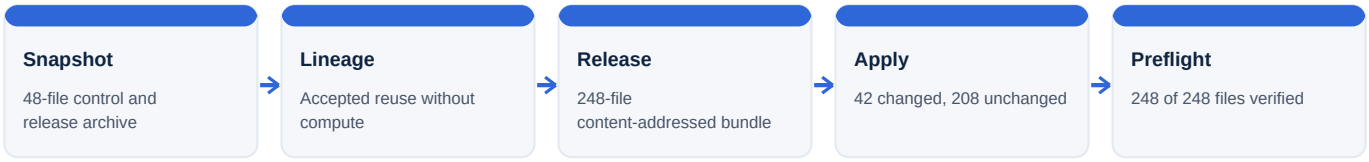
Observed five-algorithm smoke output

Logical file sizes; the safe run kept all final models and added retention evidence.

Run	Files	Logical bytes	Logical MiB	Interpretation
Original smoke 3543	57	275,209,516	262.460	Checkpoint-writing baseline
Storage-safe smoke 3550	64	137,777,145	131.395	More evidence files, about half the bytes

SECTION 2

Controlled cutover and rollback lineage



CONTROL GUARANTEES

- The original campaign specification hash was preserved, and the storage-safe specification was introduced as an explicit successor.
- The rollback archive preserves the pre-cutover campaign control plane and active release; restoring it remains an intentional operation.
- Calibration evidence from jobs 3540 and 3541 was reused without compute because its attempt, release, preflight, data, runtime, and output bindings were unchanged.
- The release build was repeated deterministically. The final bundle contains all 218 predecessor files plus 30 additions, with zero lost predecessor files.

IDENTITY LEDGER

Artifact or identifier	Verified value
Original campaign specification SHA-256	`9d0d02ad1890379e 72ce8d7513cf93dc 83a54e8afc1c6f52 7c5275e76476a0b4`
Rollback archive SHA-256	`5dd485c41ee5e287 c893081f760f1f49 6fb458e1e1097e69 1fb64e254e822aee`
Lineage receipt SHA-256	`25cda3f51aeb2003 53064e495fd58484 ade711f36ca575c9 633ae231e069c807`
Cutover release ID	`56358f2a4996f654 426814d88e622b7f 217ac89bfc02f8bb dacc27e66873b55e`
Cutover manifest SHA-256	`439b590bc4a85f03 a6d98723ee3d28e4 0acdc00d95bdfbf4 8d310f07d867e25c`
Cutover bundle SHA-256	`a5135665dd14fdfc d0a5dcdbfab70a7e 6f88fd78fc75ff94 6df9851c01a4072e`
Returned preflight SHA-256	`1ccac9827e2f0189 27d711c0fb091811 9c79f8721bd6d82d 3d08ed026ab59f0c`

SECTION 3

Remote validation and smoke evidence

Field	Verified value	Meaning
Slurm job	3550	Exactly one new HPC submission in the cutover window
State / exit	COMPLETED / 0:0	Terminal scheduler success
Elapsed	00:33:20	2,000 elapsed seconds
Allocation	2 CPU, 8G requested	compute-devel on bistka-master
Batch MaxRSS	809,680 KiB	About 790.7 MiB peak resident memory
Total CPU	32:33.567	CPU accounting captured and hashed

Algorithm	Projected full-task hours	72-hour gate	Logical smoke bytes
PPO	3.7285	PASS	4,407,179
A2C	0.2556	PASS	2,982,770
DDPG	20.4114	PASS	38,705,767
TD3	21.1114	PASS	58,017,121
SAC	21.5986	PASS	33,638,554

STRICT RETURNED-EVIDENCE GATE

Import status pass: 5 complete experiment manifests, 5 checksum-bound retention receipts, 0 checkpoint paths, and validation status pass.

EVIDENCE HASHES

Artifact or identifier	Verified value
Slurm accounting SHA-256	`6959a20d6a95d7f0 a8d57852f40302d6 6233f587d62fe9c3 29848cf2a4608601`
Smoke summary SHA-256	`e3095e1149a6cefe 9d12cef97ca38b0b 1ef1585bfa3e4d20 bf66a4ce977b88c3`
Storage validation SHA-256	`cd3cf4b65306894e 8b50068bf21ff9d6 c65ab018fb871831 7c50c858f3c5b507`
Import validation SHA-256	`dca91a1f94c4339e d524c94fa667362f 2f6aebf414ff3b99 84717e1bb77cd5fb`

FINAL EXECUTION DECISION

The cutover itself is verified. The smoke run proves the new path. Full 150-trial screening was intentionally not submitted; a later run requires a new capacity snapshot and a new bounded execution window.